

MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

**Fraud Detection and Risk Scoring System Using Apache Spark
and Machine Learning**

by

Group 1

Dương Bình An

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Phạm Đức Long

Đỗ Quốc Trung

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Nguyễn Lê Hồng Nhi

A thesis submitted in conformity with the requirements

for the degree of Master of Software Engineering

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Supervisor:

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2.

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Abstract

This thesis presents an end-to-end fraud risk scoring system built on the IEEE-CIS transaction dataset. Apache Spark performs typed ingestion, quality auditing, chronological splitting, leakage-controlled preprocessing, feature engineering, and Parquet export. A model pipeline compares logistic regression, LightGBM, XGBoost, and CatBoost using ROC-AUC and precision–recall AUC, then packages the selected model with categorical mappings and SHAP explanations. FastAPI exposes scoring, transaction exploration, imports, dataset loading, and human review, while a React dashboard supports investigation. The current official data contract reports 94 passed verification checks. The official V2 CatBoost balanced package reports validation ROC-AUC 0.9081210 and PR-AUC 0.5805993, with holdout ROC-AUC 0.8780755 and PR-AUC 0.4266435. Its isotonic calibrator and thresholds are packaged, but the promotion gate returns `not_promoted`; the configured serving version remains unchanged. Limitations include local-only MLflow tracking, duplicated threshold ownership, offline–online historical-feature mismatch, heuristic fallback risk, single-node Python training, in-memory jobs, and missing production security controls. The result is a coherent academic demonstration with explicit architecture gaps, not a production-readiness claim.

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The members of Group 1 sincerely thank their supervisor, TAN Le Duy, for the guidance and feedback provided during the development and documentation of this project.

The project team consists of Dương Bình An, Lê Quang Tuyền, Phạm Đức Long, Đỗ Quốc Trung, Dương Hồng Quân, and Nguyễn Lê Hồng Nhi. The repository records subsystem responsibilities for data processing and exploratory analysis, model development, backend engineering, and frontend engineering. The interfaces and handover contracts between these areas made it possible to evaluate the system as one end-to-end artifact.

Contents

Abstract	3
Acknowledgments	4
List of Appendices	15
1 Introduction	16
1.1 Problem context	16
1.2 Objectives	16
1.3 Research and engineering questions	17
1.4 Contributions	17
1.5 Evaluation basis	18
1.6 Scope and evidence	18
1.7 Thesis organization	19
2 Background and Related Work	20
2.1 Transaction fraud as imbalanced learning	20
2.2 The IEEE-CIS dataset	20
2.3 Apache Spark for batch processing	21
2.4 Gradient-boosted decision trees	21
2.5 Explainability with SHAP	22

2.6	Service and interface technologies	22
2.7	Related engineering concerns	22
3	Requirements and System Architecture	24
3.1	Functional requirements	24
3.2	Quality requirements	25
3.3	Integrated architecture	25
3.4	Component responsibilities	27
3.4.1	Data pipeline	27
3.4.2	Model package	27
3.4.3	Backend	27
3.4.4	Frontend	27
3.5	Primary data flows	28
3.5.1	Batch preparation	28
3.5.2	Training	28
3.5.3	Real-time scoring	28
3.5.4	Review	28
3.6	Contracts	28
3.7	Risk and trust boundaries	29
3.8	Architectural tradeoffs	30
4	Big-Data Processing and Feature Engineering	31
4.1	Source validation and typed ingestion	31

4.2	Join integrity	31
4.3	Exploratory analysis and quality controls	32
4.4	Leakage-controlled transformation order	32
4.5	Chronological split	34
4.6	Feature engineering	35
4.6.1	Time features	35
4.6.2	Amount features	35
4.6.3	Missingness and presence	35
4.6.4	Anonymized behavior fields	35
4.6.5	Historical entity features	35
4.6.6	Categorical features	36
4.7	Missing-value and class-imbalance strategies	36
4.8	Output contract	37
4.9	Verification result	37
4.10	Data limitations	37
5	Model Development, Evaluation, and Serving	39
5.1	Model-development protocol	39
5.2	Candidate models and selection	41
5.3	Temporal development windows	41
5.4	Official V2 results	42
5.5	Calibration and threshold policy	43

5.6	Packaging, checksum, and promotion	43
5.7	Request-time scoring and explanations	46
5.8	Serving configuration and MLflow	46
5.9	Model-readiness decision	46
6	Application and Deployment	48
6.1	Backend application	48
6.1.1	Persistence model	48
6.1.2	API surface	48
6.1.3	Transaction queries	49
6.1.4	CSV import	49
6.1.5	Dataset loading and jobs	50
6.2	Frontend application	50
6.2.1	Routes and workflows	50
6.2.2	Risk and explanation presentation	51
6.2.3	Review experience	51
6.2.4	Rule-based assistant	51
6.2.5	Accessibility and responsive design	51
6.3	Deployment	51
6.3.1	Training profile	52
6.3.2	Configuration	52
6.4	Application limitations	53

7	Validation and Whole-System Review	54
7.1	Validation strategy	54
7.2	Processed-data validation	54
7.3	Static and runtime reports	58
7.4	Automated tests	58
7.5	Current dependency and execution constraints	59
7.6	Whole-project findings	59
7.7	Threats to validity	60
7.7.1	Construct validity	60
7.7.2	Internal validity	61
7.7.3	External validity	61
7.7.4	Reproducibility	61
7.8	Production-readiness gates	61
7.8.1	Gate 1: reproducible serving	61
7.8.2	Gate 2: approved operating policy	62
7.8.3	Gate 3: operational controls	62
7.8.4	Gate 4: controlled promotion	62
7.9	Overall assessment	62
8	Conclusion and Future Work	63
8.1	Answers to the research questions	63
8.1.1	RQ1: leakage-controlled data handoff	63

8.1.2	RQ2: model comparison	63
8.1.3	RQ3: Spark-free serving and explanations	63
8.1.4	RQ4: production-readiness gaps	64
8.2	Technical contributions	64
8.3	Immediate future work	64
8.4	Model-development future work	65
8.5	Platform future work	65
8.6	Final statement	66
References		67
A API Reference		69
A.1	Common enums	69
A.2	Core response fields	69
A.3	Endpoint summary	70
B Feature and Risk Contracts		71
B.1	Canonical feature groups	71
B.2	Forbidden model columns	71
B.3	Risk bands	72
C Reproducible Runbook		73
C.1	Raw data placement	73
C.2	Preprocessing	73

C.3	Model training	73
C.4	Application	74
C.5	Local development	74
C.6	Rollback	75
D	Test and Contribution Matrix	76
D.1	Recommended validation commands	76
D.2	Acceptance scenarios	76
D.3	Team roster and repository role mapping	77

List of Tables

3.1	Functional requirements and implementation status	24
3.2	Quality requirements and assessment	25
3.3	Major architectural tradeoffs	30
4.1	Observed source files	31
4.2	Transaction–identity join audit	32
4.3	Chronological labeled splits	34
4.4	Training and evaluation class distributions	36
5.1	Selected official V2 training evidence	41
5.2	Official V2 validation and holdout metrics	42
5.3	Official V2 policy artifact	43
5.4	Official V2 promotion gate	44
6.1	Implemented API endpoints	49
6.2	Frontend routes	50
6.3	Principal environment variables	52
7.1	Automated and manual test coverage	58
7.2	Whole-project severity matrix	60
A.1	Core API fields	69

B.1	Canonical feature groups	71
D.1	Acceptance scenarios	77
D.2	Group 1 roster and recorded subsystem contributions	77

List of Figures

3.1	Overall system architecture	25
3.2	Layered data architecture	26
3.3	Data, feature, and evaluation contracts	29
4.1	Detailed data processing pipeline	33
4.2	Leakage-controlled transformation flow	34
5.1	Model training lifecycle	40
5.2	Temporal development windows	42
5.3	Promotion state machine	45
6.1	Default application deployment	52
7.1	Class distribution	55
7.2	Fraud rate by ProductCD	56
7.3	Fraud rate by identity presence	56
7.4	Fraud rate by transaction hour	57
7.5	V1 and V2 validation performance	57

List of Appendices

Appendix	Title	Page
A	API Reference	69
B	Feature and Risk Contracts	71
C	Reproducible Runbook	73
D	Test and Contribution Matrix	76

Chapter 1

Introduction

1.1 Problem context

Online transaction fraud is a highly imbalanced classification problem in which the positive class is rare, errors have asymmetric consequences, and the data distribution can change over time. A useful operational system must therefore do more than maximize a single offline metric. It must preserve the temporal meaning of validation, make feature transformations reproducible, expose model version and uncertainty, support investigation, and connect predictions to a review process.

The IEEE-CIS Fraud Detection dataset was released through a Kaggle competition and contains anonymized card-not-present transaction and identity information provided by Vesta Corporation (IEEE Computational Intelligence Society, 2019; Amazon Science, 2023). Its scale and mixture of numeric, categorical, missing, and identity attributes make it a useful academic basis for studying big-data preparation and tabular machine learning.

This project addresses the following system question:

How can a reproducible big-data pipeline, an imbalanced-classification model, an explainable scoring API, and a human-review interface be integrated into one auditable fraud risk scoring demonstration?

1.2 Objectives

The project has six technical objectives:

1. ingest and validate the full IEEE-CIS transaction and identity files;

2. build leakage-controlled, chronological, model-ready datasets;
3. compare interpretable and gradient-boosted classifiers using metrics suitable for an imbalanced positive class;
4. package the selected estimator behind a stable per-transaction scoring interface with local explanations;
5. expose scoring, persistence, review, import, and dataset loading through a typed API;
6. provide a usable dashboard and reproducible local deployment.

1.3 Research and engineering questions

The implementation and review are organized around four questions:

- RQ1:** Can the raw data be transformed into a versioned handoff without identifier overlap or training-to-holdout leakage?
- RQ2:** Which evaluated classifier has the strongest saved validation PR-AUC, and does its final artifact retain performance on the chronological holdout?
- RQ3:** Can the batch model be served without Spark at request time while retaining categorical compatibility and local explanations?
- RQ4:** Which gaps prevent the demonstration from supporting a production-readiness claim?

1.4 Contributions

The project contributes:

- a Spark pipeline with typed ingestion, join audits, EDA, temporal splitting, training-only transform fitting, feature contracts, and a machine-readable verifier;
- weighted and deterministic undersampling strategies that leave validation and holdout prevalence untouched;
- a versioned comparison of logistic regression, LightGBM, XGBoost, and CatBoost, with a packaged version-aware scoring interface;

- a FastAPI and SQL persistence layer for scoring, imports, dataset loads, transaction exploration, and human review;
- a React dashboard with risk visualization, SHAP evidence, data loading, review operations, and fallback-aware states;
- a whole-project audit that separates verified behavior, saved artifact claims, implemented capability, and missing evidence.

1.5 Evaluation basis

The primary predictive metrics are ROC-AUC and precision–recall AUC (PR-AUC). PR curves are particularly informative when positive and negative class counts are highly skewed because precision exposes the false-positive burden among positive predictions (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

The project does not treat area metrics as a complete operational policy. Precision, recall, confusion counts, calibration, error cost, and review capacity at a selected threshold are required before a model score can control real financial decisions.

1.6 Scope and evidence

The evidence snapshot is the repository and locally available generated artifacts inspected on 31 July 2026. Source code proves that a capability is implemented; it does not by itself prove that a training or deployment run completed. Saved JSON/CSV artifacts are reported as *artifact-reported* when they could not be independently regenerated in the review environment.

The thesis excludes:

- claims about real production fraud savings;
- personal or merchant identification from anonymized fields;
- causal explanations of fraud;
- automatic deployment or online learning;
- regulatory approval of the risk-band policy.

1.7 Thesis organization

Chapter 2 introduces the dataset, large-scale processing, boosted-tree models, evaluation, and SHAP. Chapter 3 defines requirements and the integrated architecture. Chapter 4 examines the data pipeline and its leakage controls. Chapter 5 presents model training, results, serving, and limitations. Chapter 6 documents the API, database, frontend, and deployment. Chapter 7 consolidates validation evidence and the production-readiness review. Chapter 8 answers the research questions and identifies future work.

Chapter 2

Background and Related Work

2.1 Transaction fraud as imbalanced learning

Fraud detection differs from a balanced classroom classification task. Fraud is rare, labels can be delayed, behavior changes, and the cost of a missed fraud case may differ substantially from the cost of reviewing a legitimate transaction. Accuracy can therefore be misleading: a classifier that predicts the majority class can appear accurate while detecting no fraud.

For a positive fraud class, precision and recall are:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (2.1)$$

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (2.2)$$

ROC-AUC measures ranking across true- and false-positive rates, while PR-AUC summarizes the precision–recall tradeoff. Davis and Goadrich showed the relationship between ROC and PR spaces and highlighted the usefulness of PR curves under class skew (Davis and Goadrich, 2006). Saito and Rehmsmeier further demonstrated why PR plots can be more informative for imbalanced binary classifiers (Saito and Rehmsmeier, 2015).

2.2 The IEEE-CIS dataset

The source consists of transaction tables and smaller optional identity tables. The public columns include transaction amount, product and card categories, email domains, device attributes, time, and many anonymized groups such as C, D, M, and V (IEEE Computational

Intelligence Society, 2019; Amazon Science, 2023). The training transaction file contains 590,540 rows in the inspected source, and the natural fraud prevalence is approximately 3.5%.

The project uses a left join because identity information is unavailable for most transactions. Missingness and identity presence can themselves be useful signals, but they must be interpreted carefully: association does not show that missing data causes fraud.

2.3 Apache Spark for batch processing

Apache Spark provides a unified engine and higher-level APIs for distributed data processing (Zaharia et al., 2016). This project uses Spark for typed CSV ingestion, joins, profiling, feature computation, chronological splitting, Parquet export, and a small decision-tree demonstration.

Spark is not used for request-time inference. Model training crosses from Spark to pandas because the compared Python estimators use in-memory array/data-frame interfaces. This boundary simplifies integration but constrains training scale to available driver memory.

2.4 Gradient-boosted decision trees

Gradient boosting builds an additive ensemble of decision trees, where later trees correct residual errors of the current ensemble (Friedman, 2001). Three modern implementations are compared:

- XGBoost** provides scalable tree boosting with sparsity-aware learning, weighted quantile sketches, cache-aware access, and distributed execution (Chen and Guestrin, 2016).
- LightGBM** uses histogram-based learning together with techniques such as gradient-based one-side sampling and exclusive feature bundling (Ke et al., 2017).
- CatBoost** introduces ordered boosting and categorical-statistic techniques designed to reduce prediction shift and target leakage (Prokhorenkova et al., 2018).

Logistic regression provides a lower-complexity baseline. The comparison uses the same training/validation interface, but the final V1/V2 comparison is not a controlled algorithm-only ex-

periment because feature contracts differ.

2.5 Explainability with SHAP

SHAP is a framework for additive feature attribution based on Shapley values (Lundberg and Lee, 2017). For a prediction $f(x)$, an additive explanation approximates the output as:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i, \quad (2.3)$$

where ϕ_i is the contribution attributed to feature i . The project uses a tree-specific explainer and returns the five largest absolute local contributions.

SHAP values provide a model explanation, not a causal explanation. They can also explain a different output scale from the probability shown to a user when a separate calibrator transforms the raw model score.

2.6 Service and interface technologies

FastAPI uses Python type annotations and Pydantic models to create runtime validation and OpenAPI schemas (Ramírez). SQLAlchemy maps the transaction and review entities to PostgreSQL or SQLite (Bayer; PostgreSQL Global Development Group). React provides the component model for the browser application (Meta Open Source), while Vite builds and serves the TypeScript bundle (You and Contributors).

The technologies are connected as a modular monorepo rather than independent microservices. This is appropriate for a short project because it reduces deployment complexity, but the backend image inherits heavyweight training dependencies from the model package.

2.7 Related engineering concerns

A production fraud platform would additionally require:

- a feature store or request-time method that exactly reproduces historical aggregates;
- monitored calibration and drift;
- business-approved operating thresholds;
- immutable model lineage and controlled promotion;
- durable case management and review audit history;
- authentication, authorization, secrets, and privacy governance.

The present work evaluates these as readiness gaps instead of claiming that the academic demonstration already provides them.

Chapter 3

Requirements and System Architecture

3.1 Functional requirements

The implemented system satisfies the following demonstration requirements:

Table 3.1: Functional requirements and implementation status

ID	Requirement	Status
FR-01	Validate and process the required transaction and identity CSV files.	Implemented
FR-02	Produce train, validation, holdout, and unlabeled model-ready datasets.	Verified
FR-03	Preserve a canonical feature order and preprocessing metadata.	Verified
FR-04	Compare multiple binary classifiers using validation metrics.	Artifact-reported
FR-05	Score one transaction and return probability, risk, version, and explanation.	Implemented
FR-06	Persist scored transactions and human review outcomes.	Implemented
FR-07	Filter, sort, paginate, and inspect transactions.	Tested in API suite
FR-08	Import CSV and load processed datasets with progress reporting.	
FR-09	Support review approval/rejection and a fraud/legitimate label.	Tested in API suite
FR-10	Present risk, SHAP evidence, model state, and fallback state in a browser.	Implemented

3.2 Quality requirements

Table 3.2: Quality requirements and assessment

ID	Requirement	Assessment
QR-01	Temporal leakage control and non-overlapping splits.	Verified
QR-02	Reproducible schema and feature contracts.	Verified
QR-03	Version-aware model artifacts and rollback.	Implemented
QR-04	Explainable tree-model predictions.	Implemented
QR-05	Local deployment with bounded data mounts.	Implemented
QR-06	Automated subsystem tests.	Partial
QR-07	Auditable experiment and promotion lineage.	Incomplete
QR-08	Durable background operations.	Incomplete
QR-09	Authentication, authorization, and audit controls.	Not implemented
QR-10	Production monitoring and service objectives.	Not implemented

3.3 Integrated architecture

Figure ?? shows the implemented batch and serving layers. The central contract is the verified model-ready dataset plus its feature and schema metadata.

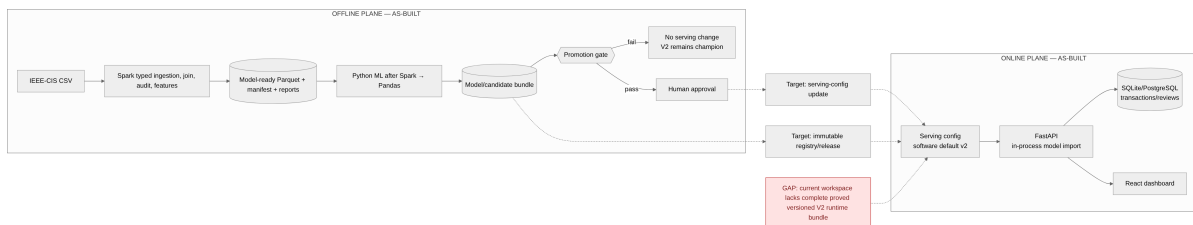


Figure 3.1: Overall system architecture (HYBRID). Solid lines are implemented and evidenced; dashed lines are proposed or partially enforced. The failed-candidate path leaves serving unchanged and retains the configured champion.

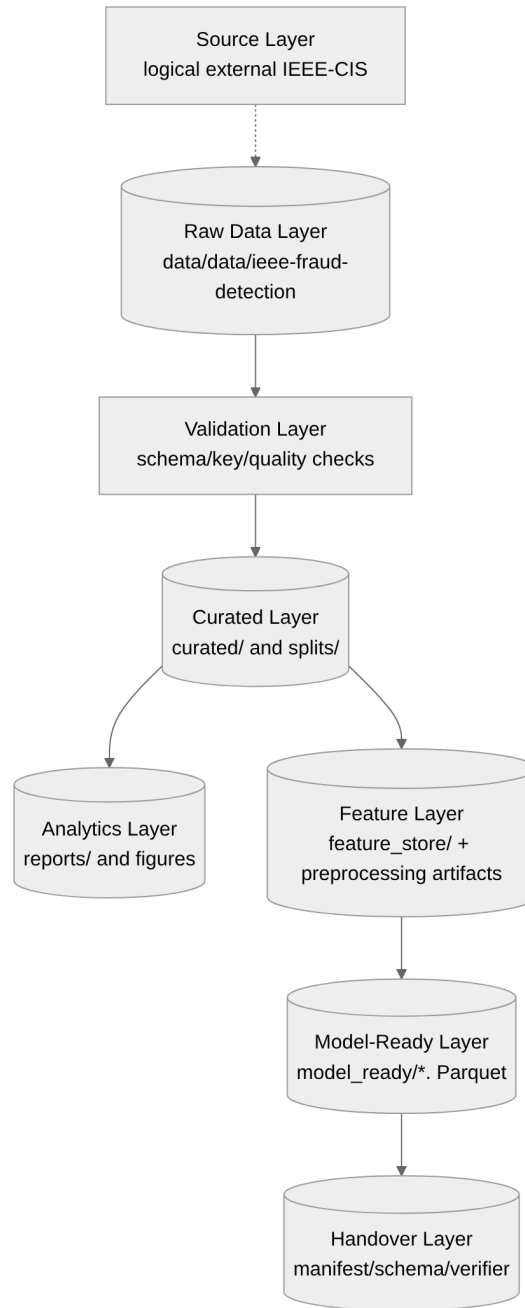


Figure 3.2: Layered data architecture (AS-BUILT with explicit logical/physical labels). Persisted paths are shown only where repository artifacts provide evidence.

The architecture uses a direct Python dependency between model and backend. This avoids network serialization and a separate model service for the demonstration. It also means backend dependency resolution and image size are coupled to the training project.

3.4 Component responsibilities

3.4.1 Data pipeline

The data component owns raw-file validation, typed ingestion, transaction–identity join preservation, quality and EDA reports, feature engineering, split policy, training-fitted transformations, Parquet output, and the manifest/verifier.

3.4.2 Model package

The model component owns dataset loading, training-only categorical indexing, candidate comparison, tuning, evaluation, artifact packaging, version path selection, SHAP initialization, and the `score(features)` interface.

3.4.3 Backend

The backend owns API validation, risk bands, automatic decisions, model/fallback state, SQL persistence, transaction queries, reviews, CSV import, processed dataset discovery, and background loading jobs.

3.4.4 Frontend

The frontend owns user navigation, filtering and pagination, risk presentation, SHAP visualization and accessible fallback, review input, ad-hoc scoring, data loading/import, polling, theme preferences, and rule-based assistance.

3.5 Primary data flows

3.5.1 Batch preparation

Raw CSV files are mounted into a Linux Spark container. The pipeline writes ignored local artifacts under `data/processed/ieee_cis_fraud_risk`. Downstream modules consume Parquet and JSON contracts, not in-memory Spark objects.

3.5.2 Training

The model loads the weighted training split and untouched validation/holdout splits, fits categorical mappings on training, converts the model matrix to pandas, selects a candidate on validation, and writes a versioned artifact.

3.5.3 Real-time scoring

The backend calls the in-process model package. Raw string categorical values are mapped using dictionaries persisted with the artifact. Probability and local SHAP evidence are mapped into risk presentation and returned through Pydantic schemas.

3.5.4 Review

Scored transactions are stored before investigation. The human review result is a separate entity so an automatic decision remains distinguishable from a reviewer's status and label.

3.6 Contracts

The system has three principal contracts:

1. the model-ready data contract, including canonical feature order and schema/version meta-

- data;
- the model scoring contract, including required/defaultable features, probability, SHAP, and version metadata;
 - the HTTP contract, including transaction, review, dataset, load, job, import, and scoring types.

The first contract is strongly machine-checked. The third is manually duplicated between Pydantic, TypeScript, mocks, and Markdown. At the evidence snapshot, the TypeScript declarations omit several backend metadata fields and `scoring_mode`; OpenAPI-generated types are recommended.

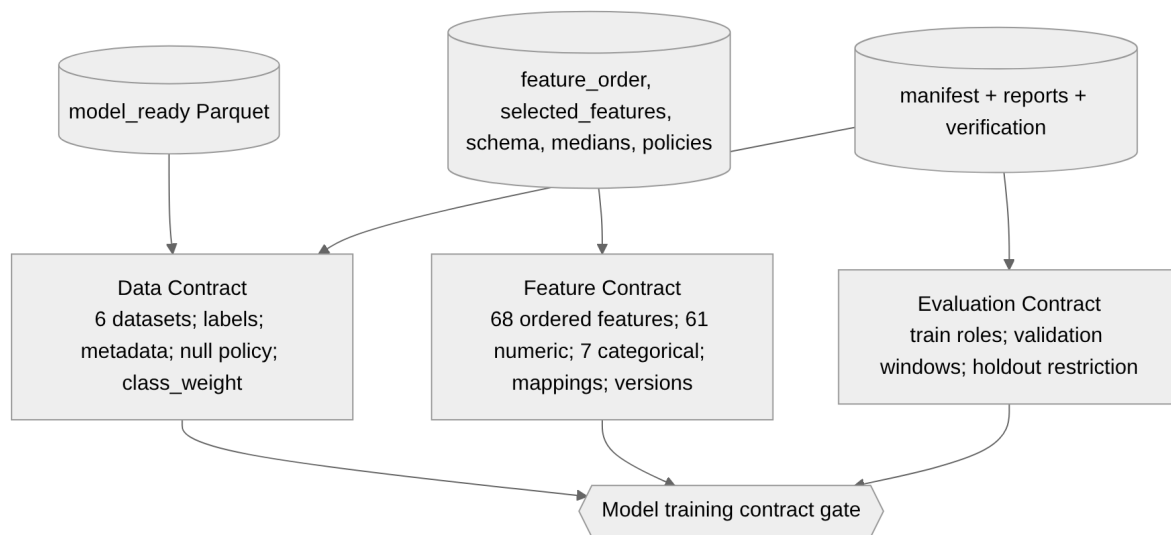


Figure 3.3: Data, feature, and evaluation contracts (AS-BUILT/HYBRID). The model-ready handover is implemented; online policy and feature-service elements shown with dashed lines remain target concerns.

3.7 Risk and trust boundaries

The browser is untrusted input. FastAPI/Pydantic validates JSON and route parameters; CSV parsing applies bounded row limits and per-row error handling. The database and processed-data mount are trusted local resources.

Joblib model artifacts are also a trust boundary because Python deserialization can execute code. Only artifacts produced by a controlled training pipeline should be loaded. The official

V2 candidate stores SHA-256 checksums, but the repository does not yet provide an external immutable registry that is the runtime serving authority.

The fallback heuristic is a further trust concern. It keeps the demonstration available when the model fails, but it must not be allowed to impersonate a production model. The backend exposes a warning, yet readiness remains fail-open. This is a degraded demo mode, not a production-equivalent score.

3.8 Architectural tradeoffs

Table 3.3: Major architectural tradeoffs

Choice	Benefit	Cost
Monorepo and path dependency	Simple handoff and one Compose stack	Backend inherits model dependencies
Spark then pandas	Scalable preparation and familiar estimators	Driver-memory training ceiling
Fixed backend risk bands	Stable UI and simple tests	Not tied to model operating evidence
In-memory jobs	Minimal infrastructure	Lost on restart and single-worker only
SQLite local default	Fast developer startup	Different operational behavior from PostgreSQL
Heuristic fallback	Demo continuity	Can conceal model loading failures
Manual TypeScript types	No generator setup	Contract drift risk

Chapter 4

Big-Data Processing and Feature Engineering

4.1 Source validation and typed ingestion

The raw inventory contains the four required IEEE-CIS files and an optional sample-submission file. The four required files occupy approximately 1.29 GB. The pipeline validates discovery, required names, sizes, headers, and schemas before performing expensive Spark actions.

Table 4.1: Observed source files

File	Bytes	Role
train_transaction.csv	683,351,067	Labeled transactions
train_identity.csv	26,529,680	Optional train identity
test_transaction.csv	613,194,934	Unlabeled transactions
test_identity.csv	25,797,161	Optional test identity

Explicit Spark schemas avoid inferring a different type across runs and make conversion failures measurable. Identity column names are normalized before the join.

4.2 Join integrity

The transaction table is the authoritative row set and identity is joined with a left join on `TransactionID`. Table 4.2 shows that the join preserves every transaction row.

Table 4.2: Transaction–identity join audit

Dataset	Transactions	Identity matches	Unmatched	Row difference
Train	590,540	144,233	446,307	0
Test	506,691	141,907	364,784	0

The large unmatched count is expected. Presence and missingness features preserve information about identity availability without dropping rows.

4.3 Exploratory analysis and quality controls

The pipeline writes source inventory, key and join audits, duplicate summaries, invalid-record summaries, missingness profiles, class distributions, temporal aggregations, product/card/device/email breakdowns, and outlier reports.

These reports are descriptive. A higher observed fraud rate for a device or product group does not establish a causal relationship; the group can be confounded by transaction mix, time, or missingness.

4.4 Leakage-controlled transformation order

Figure ?? highlights the transforms whose parameters are fitted only on the training window.

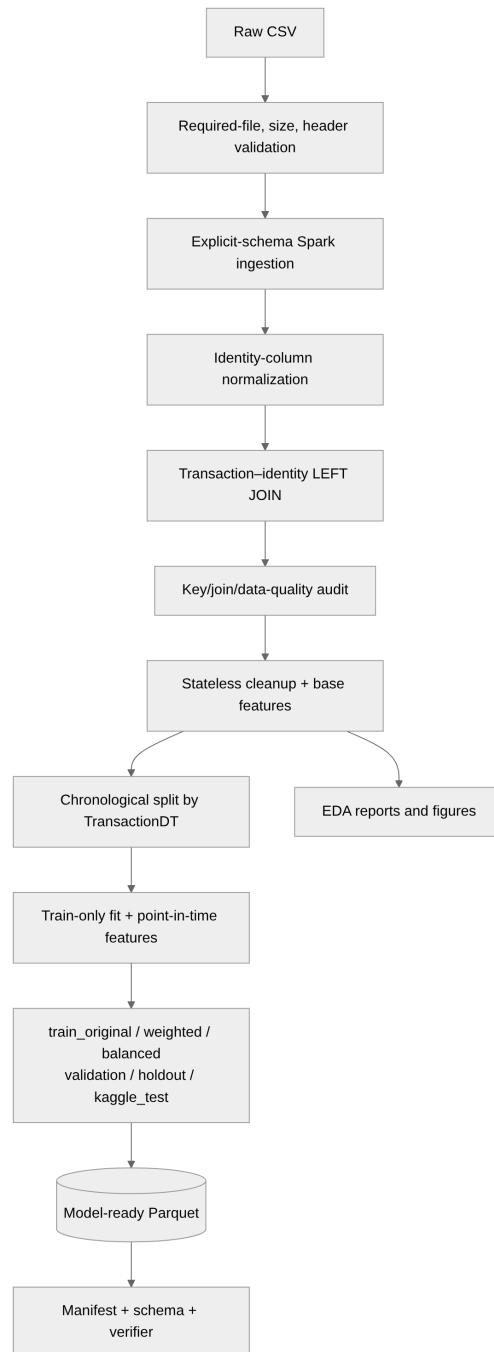


Figure 4.1: Detailed data processing pipeline (AS-BUILT). EDA is a reporting branch; it is not represented as a mandatory transformation of the model-ready data.

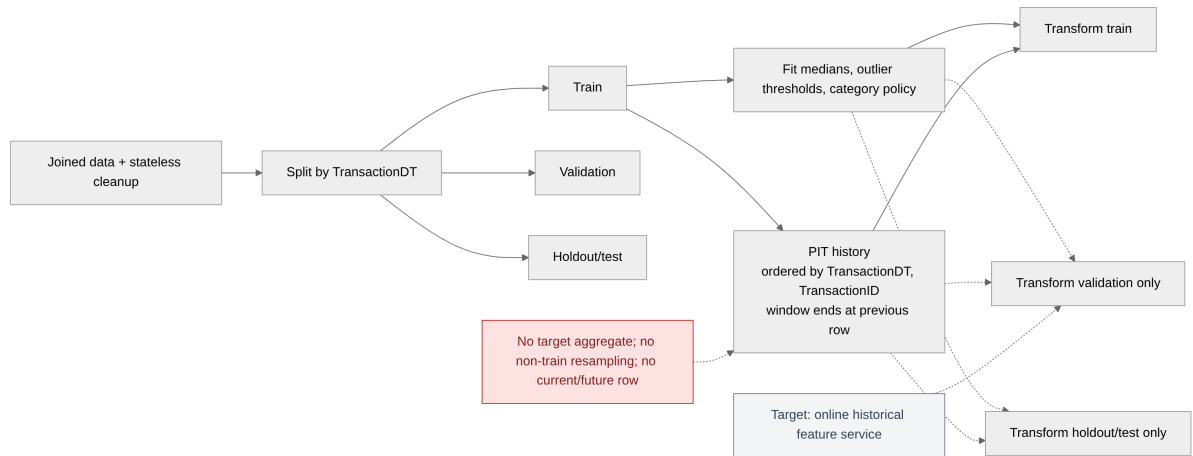


Figure 4.2: Leakage-controlled transformation flow (AS-BUILT). Learned preprocessing and historical lookups are fitted from the chronological training data and applied forward without current-row or future-row information.

The use of `TransactionDT` produces forward validation rather than a random split. This better represents the intended question: whether a model trained on earlier transactions ranks later transactions.

4.5 Chronological split

Table 4.3: Chronological labeled splits

Dataset	Rows	Fraud	Fraud rate	<code>TransactionDT</code> range
Train	412,956	14,522	3.5166%	86,400–10,432,902
Validation	88,486	3,036	3.4310%	10,432,915–13,136,583
Holdout	89,098	3,105	3.4849%	13,136,627–15,811,131

The saved verifier found unique identifiers inside each split and zero overlap between split pairs. The nearly stable prevalence across natural windows makes the comparison interpretable, although one validation window cannot quantify variation across multiple historical regimes.

4.6 Feature engineering

4.6.1 Time features

Time features include transaction day, week, hour, a day-of-week proxy, age in days, and a night-transaction flag. The raw `TransactionDT` is retained in the canonical feature order.

4.6.2 Amount features

Amount processing includes the raw value, logarithm, capped value, decimal component, amount band, and ratios/deviations relative to card history. Saved outlier thresholds are learned from training and applied forward.

4.6.3 Missingness and presence

The pipeline produces counts and ratios for selected/identity missing values and flags for identity, device, payer/recipient email, distance, address, and shared email domain presence.

4.6.4 Anonymized behavior fields

Selected `C1--C14`, `D1--D15`, and distance features are retained. Their public semantic meaning is limited, so this report describes them as behavioral/anonymized variables rather than inventing business definitions.

4.6.5 Historical entity features

Card, email, and device keys produce prior transaction counts and amount aggregates. Card history additionally includes standard deviation and time since the previous transaction. Training rows use prior-window history; later splits receive lookups fitted only from the training window.

4.6.6 Categorical features

The seven categorical fields are:

- `ProductCD`, `card4`, and `card6`;
- `DeviceType` and `device_family`;
- `M4` and `amount_band`.

String indexers are fitted on training. The extracted mappings are serialized with the model for Spark-free request-time encoding.

4.7 Missing-value and class-imbalance strategies

Numeric medians and categorical missing policies are persisted. At serving, missing values that remain after categorical mapping use a numeric sentinel. This is compatible with the current tree models, but the sentinel must remain part of the model contract.

The primary training set uses class weights and keeps all 412,956 training rows. A second set undersamples legitimate examples:

Table 4.4: Training and evaluation class distributions

Dataset	Legitimate	Fraud	Legitimate-to-fraud ratio
<code>train_original</code>	398,434	14,522	27.44
<code>train_weighted</code>	398,434	14,522	27.44
<code>train_balanced</code>	43,872	14,522	3.02
<code>validation</code>	85,454	3,036	28.15
<code>holdout</code>	85,989	3,105	27.69

Validation and holdout remain untouched. This is essential because precision and calibration depend on class prevalence.

4.8 Output contract

The pipeline exports curated data, splits, model-ready Parquet, feature-store tables, preprocessing artifacts, schema artifacts, reports, and demonstration cases. The canonical model-ready datasets are:

- `train_original`;
- `train_weighted`;
- `train_balanced`;
- `validation`;
- `holdout`;
- `kaggle_test`.

4.9 Verification result

The saved verifier reports 94 checks passed and no failures. It validates dataset presence, row counts, identifiers, labels, weights, feature order, schema hashes, chronological order, split overlap, and required artifacts.

The current official-run report records the terminal verifier result:

```
ready_for_downstream_training
```

An older persisted manifest/snapshot used `verification_pending`; it is treated as historical evidence and must not be mixed with the official-run report when generating final metrics.

4.10 Data limitations

- Feature semantics are partly anonymized.
- Entity aggregates are batch features; a production online path is not implemented.
- The training matrix is collected to pandas before estimator fitting.
- There is one chronological validation window rather than temporal cross-validation.

- Native Windows is not the canonical full Parquet execution path.
- The local processed tree is large and intentionally excluded from Git.

Chapter 5

Model Development, Evaluation, and Serving

5.1 Model-development protocol

The official V2 workflow consumes the Spark/Parquet handover and executes the following ordered stages:

1. validate the data contract and canonical feature order;
2. fit categorical mappings on training data only;
3. train a Logistic Regression baseline;
4. compare Logistic Regression, LightGBM, XGBoost, and CatBoost, including balanced variants;
5. select and tune a candidate using the selection window;
6. fit an isotonic calibrator on the calibration window;
7. choose review and reject thresholds on the policy window;
8. freeze estimator, mappings, calibrator, and policy metadata;
9. evaluate the frozen package on holdout;
10. write candidate metadata, checksums, smoke evidence, and a promotion decision.

The feature loader removes `TransactionID`, `isFraud`, `class_weight`, `split_name`, processing metadata, and `generated_at` from the model vector. The canonical vector contains 68 features, 61 numeric and 7 categorical, and includes `TransactionDT`. This prevents a previous contract failure in which the string `train_weighted` entered the numeric training frame.

The boundary must be stated precisely as: **distributed data preparation with single-node Python model training**. Spark writes Parquet; Python training converts model matrices to pandas. There is no evidence that LightGBM, XGBoost, or CatBoost is distributed over Spark

workers.

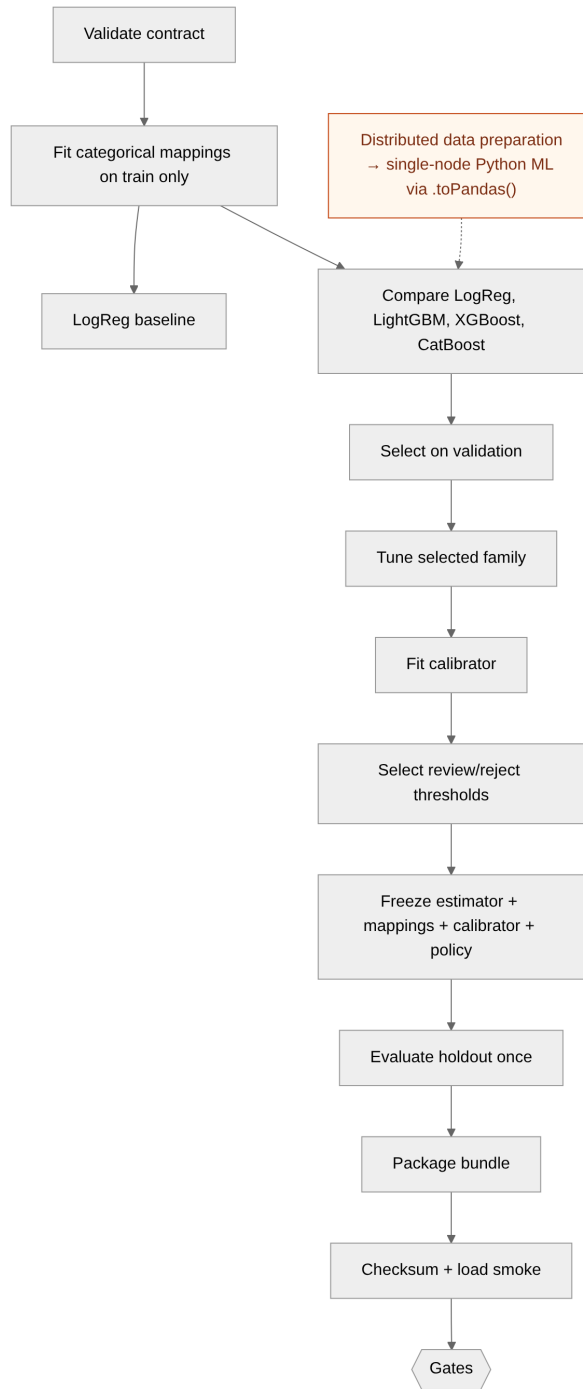


Figure 5.1: Model training lifecycle (HYBRID). Spark provides distributed data preparation; estimator fitting is single-node Python training after the Parquet-to-Pandas boundary.

5.2 Candidate models and selection

The baseline is Logistic Regression. The compared gradient-boosting families are LightGBM, XGBoost, and CatBoost. The current official V2 comparison selects `catboost__balanced`; the candidate manifest identifies the final model as CatBoost trained on the balanced variant.

Table 5.1: Selected official V2 training evidence

Field	Value
Model version	v2
Selected model	CatBoost
Training variant	balanced
Feature count	68
Selection window	44,243 rows
Calibration window	22,122 rows
Policy window	22,121 rows
Calibration method	Isotonic
Processing version	ieee-cis-preprocess-2.1.0
Feature schema	ieee-cis-features-1.1.0

The saved comparison artifact contains model-family and balanced-variant results. PR-AUC is the primary selection metric because the positive class is rare. ROC-AUC is retained as a complementary ranking metric. Accuracy is not used as a selection authority. Validation comparison is descriptive evidence; it does not establish a business operating point.

5.3 Temporal development windows

The validation period is divided into selection, calibration, and policy windows. The official V2 metadata records selection boundary 11713696 and calibration boundary 12438149. Model family and parameters are selected in the first window, isotonic calibration is fitted in the second, and thresholds are selected in the third. Holdout remains the final evaluation window and is opened after the package is frozen.

This separation is stronger than the historical V1 workflow, but it should not be used to rewrite V1’s history. If a model is modified after holdout results are viewed, the modified package must receive a new candidate identity rather than silently reusing the same holdout.

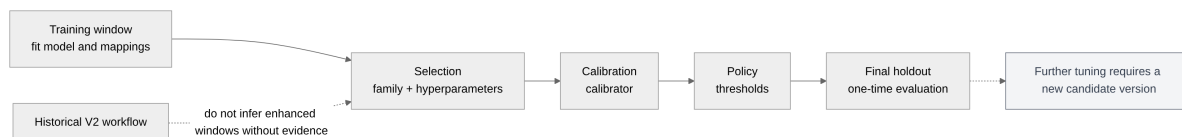


Figure 5.2: Temporal development windows (HYBRID). The enhanced candidate workflow separates selection, calibration, policy, and holdout roles; historical V2 evidence must not be rewritten as if it used every enhanced stage.

5.4 Official V2 results

Table 5.2: Official V2 validation and holdout metrics

Window	Threshold	PR-AUC	ROC-AUC	Precision	Recall	Brier
Validation selection	0.50	0.5805993	0.9081210	0.3969649	0.6220275	–
Holdout	0.16	0.4266435	0.8780755	0.3820038	0.5181965	0.0248326

The holdout confusion counts are TN 83,390, FP 2,603, FN 1,496, and TP 1,609. The raw holdout Brier score is 0.0495279 and the calibrated Brier score is 0.0248326. These numbers are artifact-reported from `reports/official-run/validation_metrics_v2.csv`, `holdout_metrics_v2.csv`, and `training_metadata_v2.json`.

The current official V2 package is not directly comparable to the historical LightGBM V1/V2 tables. The official candidate is CatBoost balanced, whereas older Markdown reports describe V2 as LightGBM and cite holdout PR-AUC 0.4582. The old values are therefore historical evidence, not current official V2 metrics. A matching V1 final holdout artifact is not present in the current repository, so this chapter does not invent a V1 holdout comparison.

5.5 Calibration and threshold policy

The candidate stores `calibration_model_v2.joblib` and records `calibration_method=isoto`

The threshold artifact selects:

Table 5.3: Official V2 policy artifact

Policy field	Value
Review threshold	0.16
Reject threshold	0.53
Minimum review precision	0.30
Minimum reject precision	0.60
Selected review precision	0.3371429
Selected reject precision	0.7397959

The model scorer applies the calibrator when the loaded artifact contains one. However, the backend runtime decision authority is `backend/src/fraud_backend/risk.py`: it rounds probability to a 0–100 score, applies fixed bands, and maps those bands to approve, review, or reject. The backend does not load the candidate review/reject thresholds as its decision policy. This is duplicated ownership, not evidence that the candidate policy is active in serving.

5.6 Packaging, checksum, and promotion

The V2 candidate package includes the final joblib model, training metadata, calibrator, threshold configuration, validation and holdout metrics, threshold analysis, candidate manifest, promotion decision, and SHA-256 checksums. The candidate manifest records artifact-load smoke passed, model name `catboost`, feature count 68, processing/schema versions, MLflow run ID, Git commit, and `git_dirty=true`.

Table 5.4: Official V2 promotion gate

Gate	Result	Evidence
Holdout PR-AUC ≥ 0.4582	Fail	0.4266435
Holdout ROC-AUC ≥ 0.8746	Pass	0.8780755
Review precision ≥ 0.30	Pass	0.3820038
Calibration non-regression	Pass	Brier decreased
Artifact checksum	Pass	SHA-256 match
Artifact-load smoke	Pass	recorded in manifest
Data contract verification	Pass	ready, 94 checks
Version match	Pass	v2
Git clean	Fail	git_dirty=true

The resulting decision is `not_promoted`, with `serving_version_unchanged=true`. This is correct model governance: the failed candidate does not transition into V2; the serving configuration remains unchanged.

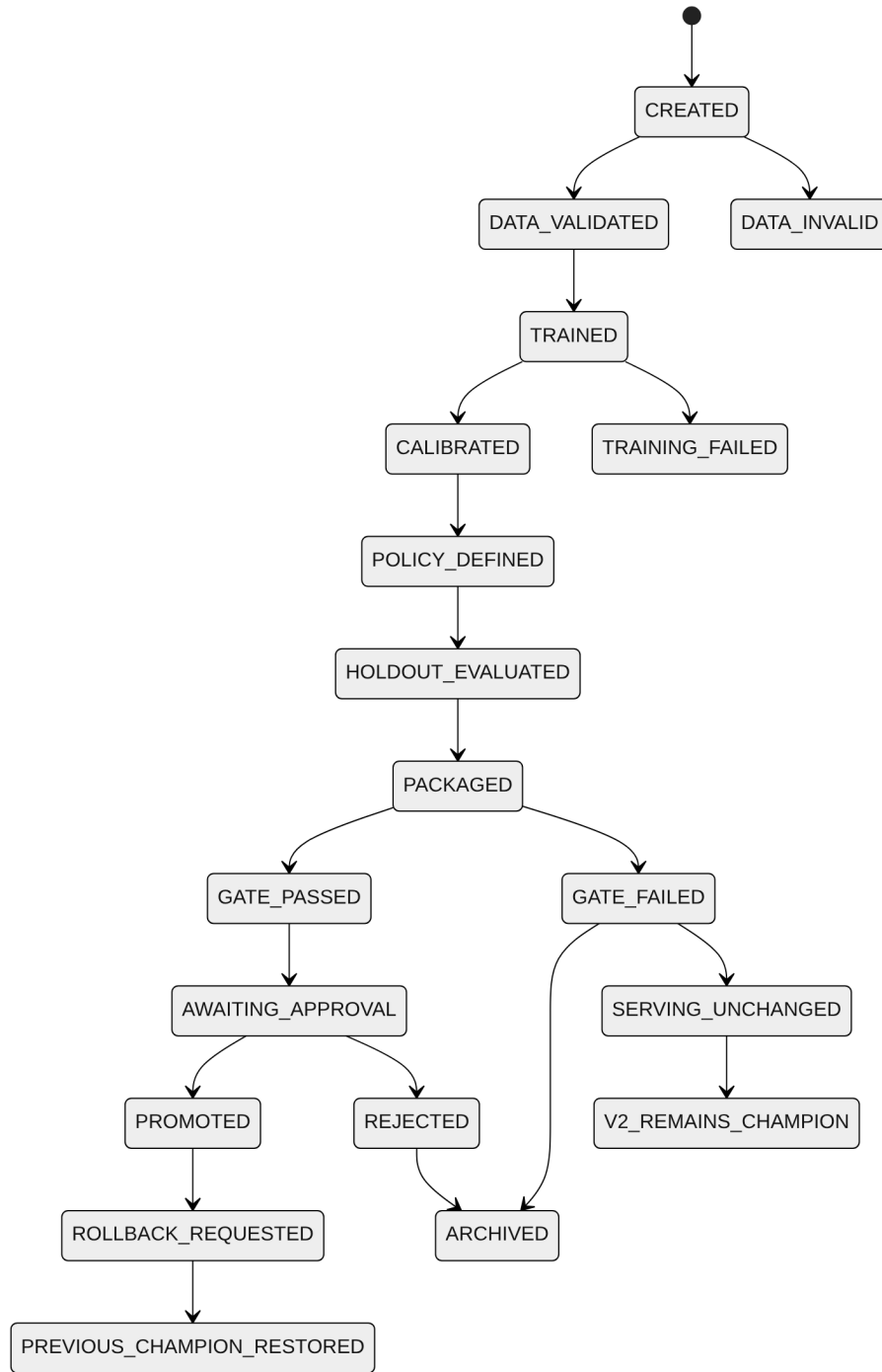


Figure 5.3: Promotion state machine (HYBRID). A failed candidate enters a failure/archive path; it does not transition into V2. Serving configuration remains unchanged until an approved update is made.

5.7 Request-time scoring and explanations

The scorer loads one artifact into a process cache and maps raw categorical strings using training-derived dictionaries. It aligns columns to the canonical order, converts missing values to the configured sentinel, calls `predict_proba`, applies an available calibrator, and returns the top five absolute TreeSHAP contributions for supported tree models. SHAP is a local model attribution and not a causal explanation. When calibration changes the displayed probability, the current SHAP values still describe the raw model output unless a separate calibrated explanation method is used.

The response distinguishes `full_feature` from `partial_demo`. Full-feature scoring requires the canonical engineered vector. Partial-demo scoring defaults missing fields and is useful for UI continuity, but it cannot recreate card, email, and device histories from a small request. It must not be described as real-time full-feature scoring.

5.8 Serving configuration and MLflow

Configuration defaults `FRAUD_MODEL_SERVING_VERSION=v2`, while the official promotion decision remains `not_promoted`. The safe wording is “v2 configured software default”, not “production-promoted model”. Setting `FRAUD_MODEL_SERVING_VERSION=v1` remains the rollback/configuration path, subject to artifact compatibility.

MLflow is implemented as local experiment tracking and lineage using SQLite and local artifact storage. The repository does not evidence a central Model Registry, registered model versions, stage transitions, or an automatic promotion authority. Candidate packaging and promotion JSON are evidence files, not a serving registry.

5.9 Model-readiness decision

The official V2 package is suitable for continued academic evaluation and manual review of its evidence. It should not be described as a production model until a frozen application smoke loads the expected artifact without fallback, the runtime policy is explicitly approved and aligned with the candidate, a durable registry/checksum handover exists, and online historical feature

parity is addressed.

Chapter 6

Application and Deployment

6.1 Backend application

The FastAPI application creates SQL tables during startup, loads the scorer, configures CORS, and registers transaction and data/job routers. PostgreSQL is used in Compose; SQLite is the default for local development.

6.1.1 Persistence model

The `Transaction` entity stores:

- transaction identifier and amount;
- raw feature dictionary;
- probability, score, band, and automatic decision;
- SHAP top-five payload;
- model version and score timestamp.

The `Review` entity stores status, fraud/legitimate label, reviewer, note, and update time. A missing review is exposed as `pending`. A subsequent review replaces the existing review for the transaction.

6.1.2 API surface

Table 6.1: Implemented API endpoints

Method	Path	Purpose
GET	/health	Process health
GET	/meta	Model/schema/feature/band metadata
POST	/score	Non-persistent ad-hoc scoring
GET	/transactions	Filtered, sorted, paginated list
GET	/transactions/stats	Aggregate dashboard statistics
GET	/transactions/import/template	Model-aligned CSV template
POST	/transactions/import	Bounded synchronous CSV scoring/import
GET	/transactions/{id}	Transaction detail
POST	/transactions/{id}/review	Human decision and label
GET	/datasets	Available processed datasets
POST	/data/load	Start a background dataset load
GET	/jobs/{id}	Job status
GET	/jobs/active/current	Active job or null
POST	/jobs/{id}/cancel	Cooperative cancellation

The exact request/response types are reproduced in Appendix A.

6.1.3 Transaction queries

List queries support exact risk/decision/review filters, inclusive score range, exact numeric identifier search, four sort modes, and bounded pagination. SQL aggregates calculate statistics without loading the full table into Python. Every risk band is represented in the statistics response, including zero-count bands.

6.1.4 CSV import

The import route requires a UTF-8 CSV with `TransactionID`. A per-row SQL savepoint means one bad row does not roll back the full batch. Coverage diagnostics report how many model columns are present. This is important because raw Kaggle files do not include every

engineered feature.

6.1.5 Dataset loading and jobs

The dataset route discovers model-ready Parquet. A load can use:

- head:** the first bounded rows;
- sample:** a deterministic random sample;
- coverage:** a deliberately non-representative set covering all five risk bands.

Only one in-memory job is active. Status is lost after restart and is not safe across multiple Uvicorn workers.

6.2 Frontend application

6.2.1 Routes and workflows

Table 6.2: Frontend routes

Route	Workflow
/	Transaction KPIs, filtering, sorting, and pagination
/transactions/:id	Risk, features, SHAP, and review
/review	Pending medium/high/critical cases
/score	Partial ad-hoc scoring form
/import	Dataset loading, job progress, template, and CSV import

TanStack Query manages server state, polling, and invalidation (TanStack). The API abstraction can switch to local JSON mocks through `VITE_USE MOCKS=true`. Mock scores preserve UI development but are explicitly not model results.

6.2.2 Risk and explanation presentation

Risk badges combine color and text. The score bullet displays the 0–100 scale and band thresholds. The SHAP component uses diverging bars and preserves a numeric table for accessibility. A null SHAP response renders an explicit unavailable state.

6.2.3 Review experience

The review form requires both an approve/reject action and a fraud/legitimate label. Optional reviewer and note fields are bounded by backend validation. Mutations invalidate detail, list, queue, and statistic queries so the new state appears consistently.

6.2.4 Rule-based assistant

The chat dock parses a limited grammar for tasks such as finding transactions or explaining displayed evidence. It is deterministic code, not a generative AI system. The distinction is important for system scope and risk.

6.2.5 Accessibility and responsive design

The implementation includes visible labels, focus styles, reduced-motion support, semantic text beyond color, system/light/dark themes, and enlarged mobile controls. The project records a manual checklist at widths 375, 768, 1024, and 1440 pixels.

There are no automated component, accessibility, or browser end-to-end tests. TypeScript types are manually synchronized and currently lag several backend metadata fields.

6.3 Deployment

Figure 6.1 shows the default application stack.

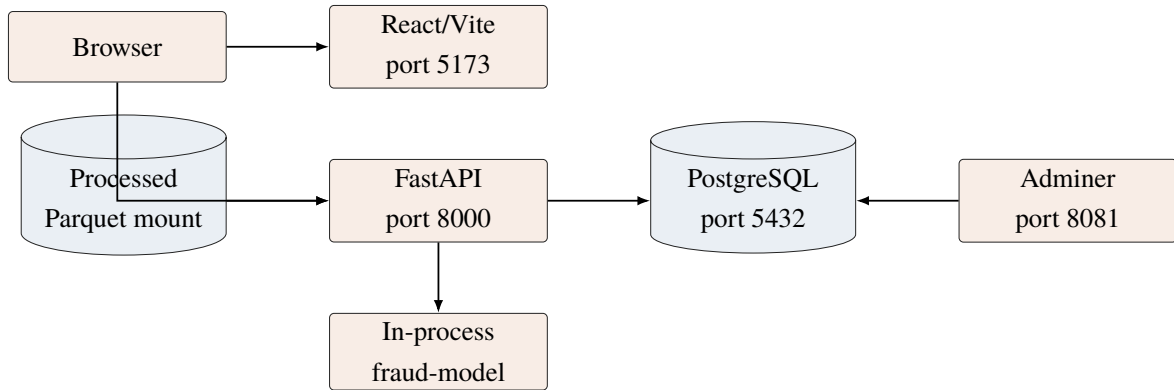


Figure 6.1: Default application deployment. Adminer is optional; Spark training services are a separate Compose profile and are not needed for request-time scoring.

The backend receives processed Parquet through a read-only mount. Model code is installed as a path dependency inside the backend image. Adminer is an optional `tools` profile.

6.3.1 Training profile

The separate `training` profile starts Spark master and worker services plus a model-training container. A local-Spark training service is available when the cluster image cannot be resolved. The Spark UI and Adminer use different host ports (8080 and 8081).

6.3.2 Configuration

Table 6.3: Principal environment variables

Variable	Purpose	Default
<code>DATABASE_URL</code>	SQLAlchemy connection	Local SQLite
<code>CORS_ORIGINS</code>	Browser origins	Local Vite origins
<code>DATA_PROCESSED_DIR</code>	Host data mount	<code>./data/processed</code>
<code>SEED_DATASET</code>	Dataset for CLI seed	holdout
<code>SEED_LIMIT</code>	Seed rows	5,000
<code>MAX_IMPORT_ROWS</code>	CSV cap	5,000
<code>FRAUD_MODEL_SERVING_VERSION</code>	Served model	v2

Variable	Purpose	Default
FRAUD_MODEL_TRAINING_VERSION	Training target	v2
SPARK_MASTER_URL	Spark target	Local/profile
MLFLOW_TRACKING_URI	Tracking backend	Local unless set

6.4 Application limitations

- no authentication, authorization, rate limiting, or audit event log;
- no Alembic migrations;
- process health is not strict real-model readiness;
- job registry is in memory and single-worker;
- CSV import is synchronous;
- no bulk re-scoring after a model change;
- backend image includes training-heavy dependencies;
- no automated frontend/browser test suite;
- frontend API types are manually maintained;
- default database credentials are development-only.

Chapter 7

Validation and Whole-System Review

7.1 Validation strategy

The review distinguishes five evidence states:

Verified now:	a command completed successfully during the documentation task.
Artifact-reported:	a value was read from a saved machine-readable artifact.
Previously validated:	a dated report records a successful prior run.
Implemented:	source code provides the capability but it was not exercised end to end.
Not evidenced:	neither a successful run nor adequate saved artifact was found.

This prevents the presence of a script or dependency from being reported as a completed experiment.

7.2 Processed-data validation

The data verifier is the strongest machine-readable quality gate. The current official-run report records:

- 94 checks passed and zero failed;
- model-ready datasets are non-empty;
- transaction identifiers are unique;
- labeled and unlabeled schemas use the expected target policy;
- class weights appear only in the weighted training variant;
- split order is chronological and split intersections are empty;

- manifest row counts and schema hashes match;
- feature order, medians, and required reports exist.

The official-run status is `ready_for_downstream_training`. An older persisted manifest and an older facts file used `verification_pending`; this is a snapshot discrepancy and should not be combined with the current official report.

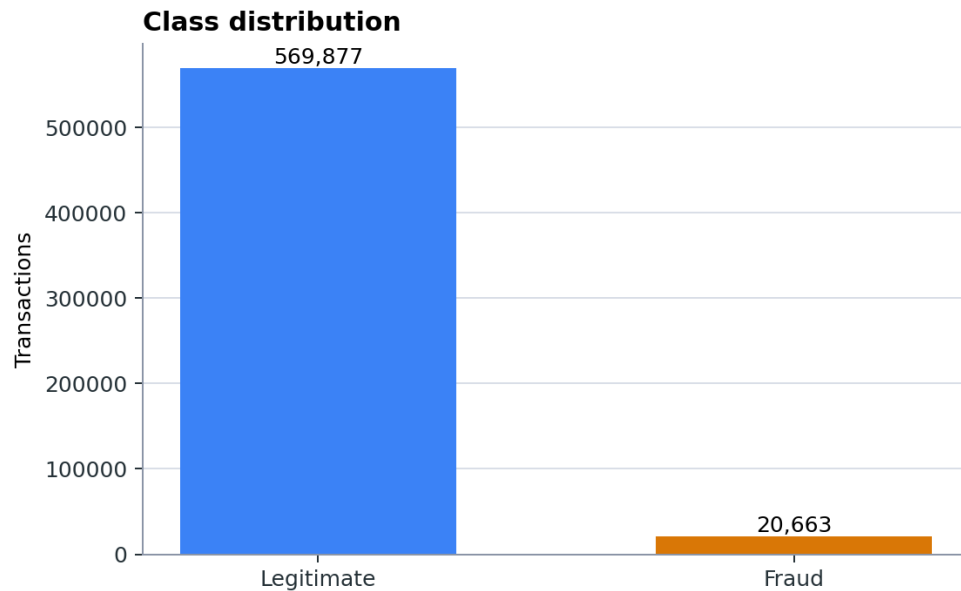


Figure 7.1: Class distribution (ARTIFACT-REPORTED). Source: official data-quality/EDA report. The chart is descriptive and does not imply causal relationships.

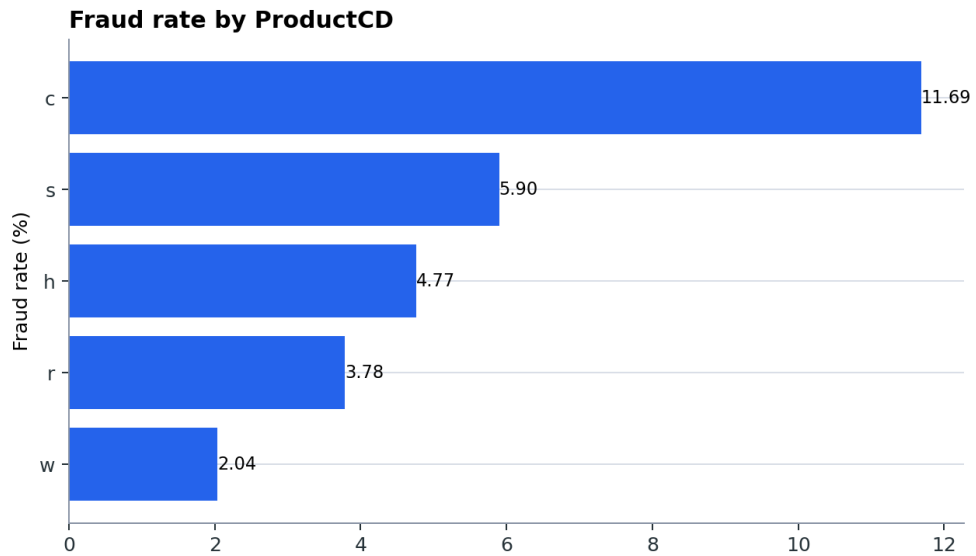


Figure 7.2: Fraud rate by ProductCD (ARTIFACT-REPORTED). Source: official EDA report; group differences are descriptive associations.

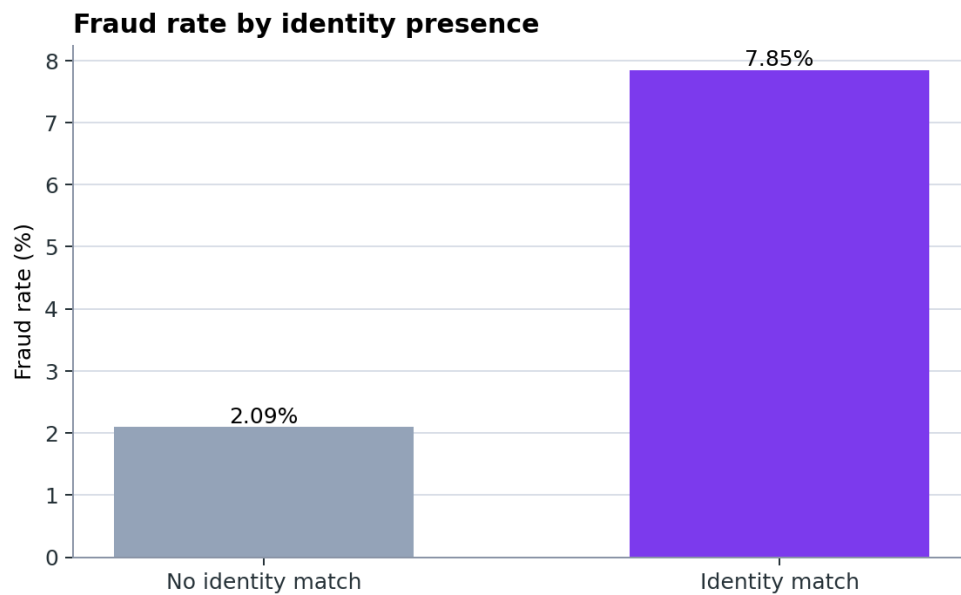


Figure 7.3: Fraud rate by identity presence (ARTIFACT-REPORTED). Source: official EDA report; identity availability is not interpreted as a causal factor.

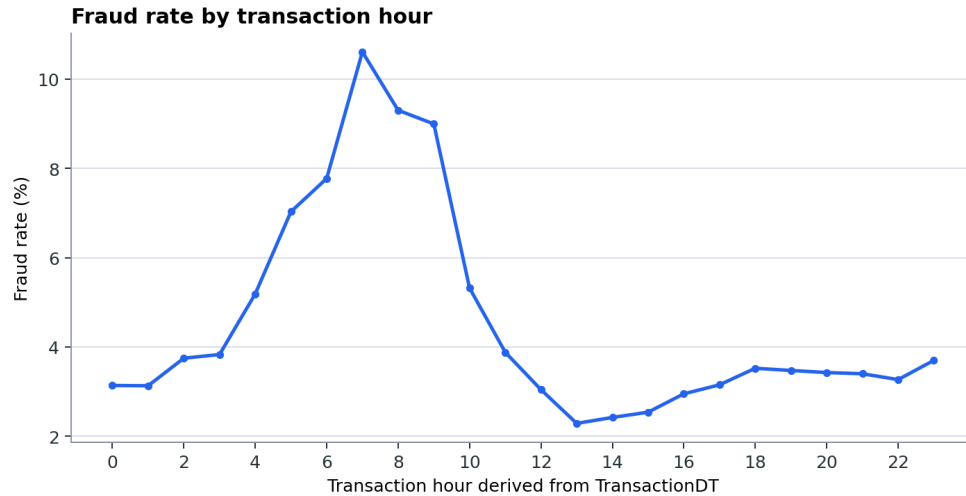


Figure 7.4: Fraud rate by transaction hour (ARTIFACT-REPORTED). Source: official EDA report; the plot summarizes temporal association only.

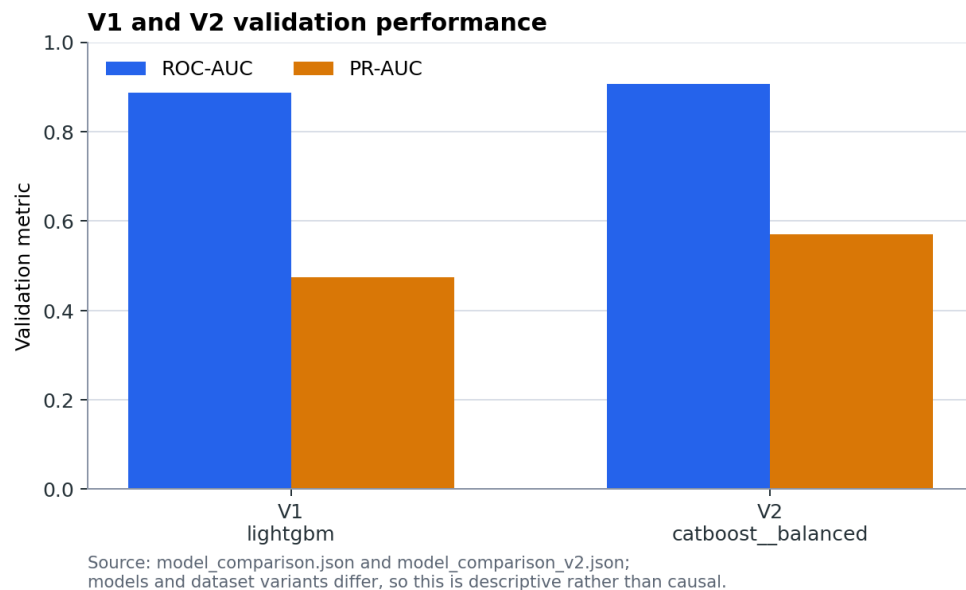


Figure 7.5: V1 and V2 validation performance (HISTORICAL/ARTIFACT-REPORTED). The comparison is retained as historical evidence and is not treated as a controlled algorithm-only ablation.

7.3 Static and runtime reports

The dated static report records successful Python compilation, YAML/JSON parsing, Compose configuration validation, leakage-pattern inspection, contract artifact presence, and Docker build-context correction.

The runtime report records the failure/remediation history rather than only the final pass:

1. Windows Spark report paths conflicted with existing directories;
2. Matplotlib attempted to use a GUI backend;
3. Docker build context excluded required source;
4. a long run left an incomplete/stale manifest;
5. normalization and contract regeneration repaired the output;
6. the verifier reached `ready_for_downstream_training`.

This history is valuable evidence because it documents why Docker Linux is the canonical Parquet path and why the contract-finalization utility exists.

7.4 Automated tests

Table 7.1: Automated and manual test coverage

Area	Covered behavior	Gap
Data verifier	Schema, rows, IDs, split order/overlap, artifacts	Full pipeline is expensive
Model features	Training-only indexer, unseen category, alignment	No multi-window validation
Model scoring	Probability, SHAP, missing values, fallback	No latency/calibration test
Model quality	Minimum holdout threshold if artifact is loadable	Can skip without artifact

Area	Covered behavior	Gap
MLflow configuration	Local-directory creation and remote-URI behavior	Full server not exercised
Backend risk	Every band boundary and 0–100 coverage	Fixed-policy only
Backend API	Health, metadata, list/stats/detail, review, score, import	Real model may be replaced
Frontend build/lint	Static typing and source lint	No component or browser tests
Frontend manual QA	Contrast, focus, motion, responsive widths	Not CI-enforced

Only model tests have a GitHub Actions workflow. A whole-system CI gate should also run backend tests, frontend lint/build, API contract generation, processed contract smoke checks, and one real-model scoring smoke.

7.5 Current dependency and execution constraints

The official model-quality log records lint success and 14 passing model tests. The stored official application-build log stopped at the frontend `npm ci` step because the lock file was out of sync with `package.json`; the missing `@emnapi` optional dependencies were then restored in the repository lock file, and a subsequent standalone frontend Docker build passed. The historical application log therefore cannot be used as evidence of a completed full-stack run.

This is a reproducibility boundary rather than a model metric failure. A new official run should preserve the repaired lock file, rebuild both images, start the application, and store health/meta/full-feature scoring responses together with the model checksum.

7.6 Whole-project findings

Table 7.2: Whole-project severity matrix

ID	Finding	Severity	Impact
R1	No authentication, authorization, or audit controls	Critical	Blocks external deployment
R2	Model readiness can fall back to heuristic	High	Predictions can be misidentified
R3	Candidate thresholds are not the backend decision authority	High	Blocks policy approval
R4	Local MLflow runs exist but central registry/stage authority is absent	High	Blocks portable lineage claim
R5	Candidate thresholds are not the served backend policy	High	Blocks operational policy claim
R6	Copied official split summary differs from current manifest split counts	Medium	Snapshot ambiguity
R7	Jobs and schema lifecycle are non-durable	Medium	Restart/upgrade risk
R8	No automated frontend/browser tests	Medium	UI regression and accessibility risk
R9	API types are manually synchronized and stale	Medium	Client contract risk
R10	Training collects full data to pandas	Medium	Driver-memory ceiling
R11	No bulk re-score/version-isolated views	Medium	Mixed-version analytics

7.7 Threats to validity

7.7.1 Construct validity

ROC-AUC and PR-AUC measure ranking, not business value. The saved results do not establish fraud prevented, investigation effort, or customer impact.

7.7.2 Internal validity

V1 and V2 differ in feature count and preprocessing controls. Their comparison cannot isolate the cause of improvement. A single validation window can also be overfit by repeated model and parameter selection.

7.7.3 External validity

IEEE-CIS is anonymized historical competition data. Behavior, prevalence, labels, feature availability, and latency constraints can differ from a live merchant environment.

7.7.4 Reproducibility

Processed data is locally present and well contracted, and the official V2 candidate includes SHA-256 checksums and one MLflow holdout run ID. MLflow remains local SQLite tracking with container-local artifact paths; a central registry ID, human approval record, and rollback smoke evidence are not present.

7.8 Production-readiness gates

7.8.1 Gate 1: reproducible serving

1. freeze dependencies;
2. build the backend/model image;
3. load V2 with the expected model version;
4. score a full-feature holdout transaction without fallback;
5. record image digest and model checksum.

7.8.2 Gate 2: approved operating policy

1. reproduce the official V2 calibration and threshold evidence on temporal windows not reused for model selection;
2. compare the candidate operating points with the configured backend policy, including precision, recall, false positives, false negatives, and calibration;
3. define review capacity and error cost;
4. approve, store, and enforce review/reject thresholds;
5. test exact boundaries through model, backend, and UI.

7.8.3 Gate 3: operational controls

1. fail readiness when the expected model is unavailable;
2. add structured logs, metrics, drift monitoring, and alerting;
3. add migrations and durable job orchestration;
4. add authentication, authorization, audit history, and secret management;
5. add CI across data, model, backend, frontend, and full-stack smoke.

7.8.4 Gate 4: controlled promotion

Run shadow scoring, compare paired predictions, approve the version with a recorded decision, test rollback, and prevent mixed-version historical statistics from being interpreted without a model-version filter or re-score.

7.9 Overall assessment

The system is successful against its academic demonstration objective. All major layers exist and the data handoff is strongly verified. The gap to production is not another dashboard page or another estimator. It is the set of controls that makes an existing prediction trustworthy: reproducibility, operating policy, lineage, readiness, security, and monitoring.

Chapter 8

Conclusion and Future Work

8.1 Answers to the research questions

8.1.1 RQ1: leakage-controlled data handoff

The project produces chronological, non-overlapping model-ready splits and persists the transformation artifacts needed by downstream training. The saved verifier reports 94 passed checks and no failures in the current official run. The official report is ready for downstream training; an older manifest/facts snapshot used `verification_pending` and is treated as historical.

8.1.2 RQ2: model comparison

The current official V2 comparison selects CatBoost on the balanced training variant. Its validation ROC-AUC is 0.9081210 and PR-AUC is 0.5805993; holdout ROC-AUC is 0.8780755 and PR-AUC is 0.4266435. The package includes isotonic calibration, thresholds, checksums, and smoke evidence, but the promotion gate returns `not_promoted` because holdout PR-AUC is below the 0.4582 reference and the Git state is dirty. V2 remains the configured software default; this is not a production approval claim.

8.1.3 RQ3: Spark-free serving and explanations

The model package serializes categorical mappings and feature order so FastAPI can score without Spark. Supported tree artifacts return local SHAP contributions. This design successfully separates batch feature preparation from request-time inference, subject to the availability of the same engineered features in a real online environment.

8.1.4 RQ4: production-readiness gaps

The principal gaps are:

- local-only experiment history without a central registry/stage authority;
- candidate calibration and thresholds are not the served backend policy;
- fail-open heuristic fallback;
- no authentication, durable jobs, or migrations;
- no drift/latency/service monitoring;
- manual client contract synchronization;
- split-count snapshot mismatch and an incomplete stored full-stack run;
- uneven CI and frontend test coverage.

8.2 Technical contributions

The system demonstrates a complete flow from multi-gigabyte source data to a human review decision:

1. Spark performs typed ingestion, audits, EDA, chronological splitting, and leakage-controlled feature engineering.
2. Explicit contracts make the data–model handoff inspectable.
3. Multiple classifiers are compared under an imbalanced metric.
4. The selected estimator is packaged with categorical mappings and SHAP.
5. FastAPI, SQLAlchemy, and React implement an explainable review workflow.
6. Docker Compose provides local application and training profiles.

8.3 Immediate future work

The first release-hardening milestone should not train a new model. It should make V2 independently reproducible and observable:

1. complete dependency locking and a frozen container smoke;
2. record model checksum, image digest, data/schema versions, MLflow run, and promotion decision in a durable registry;
3. reproduce the official V2 application smoke and operating evidence on independent windows;
4. make fallback an explicit demo mode and enforce readiness;
5. synchronize/generated API client types and expand CI.

8.4 Model-development future work

After reproducibility:

- perform temporal cross-validation;
- separate model selection, calibration, threshold selection, and holdout;
- run feature-group ablations;
- evaluate paired V1/V2 errors by segment;
- measure inference latency and resource consumption;
- monitor delayed-label performance and drift.

8.5 Platform future work

A deployable platform requires:

- authentication, authorization, secrets, and immutable audit events;
- Alembic migrations and durable background queues;
- a serving-only model dependency profile;
- production metrics, logs, tracing, readiness, and alerts;
- bulk re-scoring or version-isolated reporting;
- retention, privacy, and incident-response policies.

8.6 Final statement

The project is a coherent and well-integrated academic fraud risk scoring demonstration. Its strongest contribution is not a single metric; it is the connection of data contracts, a trained model, explanations, API behavior, persistence, and human review. The whole-project audit defines the additional evidence and controls required to turn that demonstration into a trustworthy deployment candidate.

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Chapter A

API Reference

A.1 Common enums

```
RiskBand = low | guarded | medium | high | critical
Decision = approve | review | reject
ReviewState = pending | approved | rejected
ReviewLabel = fraud | legit
ScoringMode = full_feature | partial_demo
```

A.2 Core response fields

Table A.1: Core API fields

Field	Type	Meaning
fraud_probability	float	Model/fallback output in $[0, 1]$
risk_score	integer	Rounded probability times 100
risk_band	enum	Fixed backend score band
decision	enum	Automatic backend policy result
scoring_mode	enum	Full vector or partial demonstration
shap_top5	list/null	Local model contributions when available
model_version	string	Artifact version used for the score
processing_version	string/null	Data processing lineage
feature_schema_version	string/null	Feature contract lineage

A.3 Endpoint summary

Method	Path	Response
GET	/health	Process status
GET	/meta	Model and feature metadata
POST	/score	Score response
GET	/transactions	Paginated transactions
GET	/transactions/stats	Aggregate statistics
GET	/transactions/import/template	UTF-8 CSV
POST	/transactions/import	Import outcome and coverage
GET	/transactions/{id}	Transaction detail
POST	/transactions/{id}/review	Updated detail
GET	/datasets	Available datasets
POST	/data/load	Background job
GET	/jobs/{id}	Job state
GET	/jobs/active/current	Job or null
POST	/jobs/{id}/cancel	Updated job

The normative, implementation-aligned contract is maintained in `docs/API_CONTRACT.md`.

Chapter B

Feature and Risk Contracts

B.1 Canonical feature groups

Table B.1: Canonical feature groups

Group	Representative features
Amount	TransactionAmt, log/capped/decimal amount, amount band, card-mean ratios and deviations
Time	TransactionDT, day, week, hour, day-of-week proxy, age, night flag
Missingness	Selected/identity missing counts and ratios
Presence	Identity, device, payer/recipient email, distance, address, same-domain flags
Anonymous behavior	dist1, dist2, selected C1--C14, selected D1--D15
Card history	Prior count/sum/mean/stddev and time since previous transaction
Email history	Prior count/sum/mean
Device history	Prior count/sum/mean
Categorical	ProductCD, card4, card6, DeviceType, device_family, M4, amount_band

B.2 Forbidden model columns

```
TransactionID
isFraud
class_weight
split_name
```

```
processing_version
feature_schema_version
generated_at
```

B.3 Risk bands

Minimum	Maximum	Band	Decision
0	19	low	approve
20	39	guarded	approve
40	59	medium	review
60	79	high	review
80	100	critical	reject

The risk bands are a fixed demonstration policy. They are not derived from the saved V2 validation operating point.

Chapter C

Reproducible Runbook

C.1 Raw data placement

```
data/data/ieee-fraud-detection/  
  train_transaction.csv  
  train_identity.csv  
  test_transaction.csv  
  test_identity.csv  
  sample_submission.csv # optional
```

C.2 Preprocessing

```
docker compose -f docker-compose.preprocessing.yml build  
docker compose -f docker-compose.preprocessing.yml run --rm  
  preprocess  
docker compose -f docker-compose.preprocessing.yml run --rm verify-  
  processed
```

If a completed Spark export lacks a finalized manifest:

```
python -m pipeline.processed_contract \  
  --output-dir data/processed/ieee_cis_fraud_risk  
python -m pipeline.verify_processed_data \  
  --output-dir data/processed/ieee_cis_fraud_risk
```

C.3 Model training

```
bash scripts/run_official_full.sh
```

The official workflow runs preprocessing, contract validation, baseline, comparison, tuning, calibration, threshold analysis, holdout evaluation, packaging, checksum generation, and the promotion gate. The current official decision is `not_promoted`; the serving configuration remains unchanged. Holdout evaluation must occur only after model and policy selection are fixed.

C.4 Application

```
docker compose up --build
```

Then open:

- dashboard: `http://localhost:5173`;
- API documentation: `http://localhost:8000/docs`;
- optional Adminer: `http://localhost:8081`; select the PostgreSQL system, then use the configured database credentials.

C.5 Local development

```
# Terminal 1
cd backend
uv sync --frozen
uv run uvicorn fraud_backend.main:app --reload

# Terminal 2
cd frontend
npm ci
npm run dev
```

C.6 Rollback

```
FRAUD_MODEL_SERVING_VERSION=v1
```

Restart the backend and verify `/meta` before accepting traffic. The rollback setting is a configuration path, not evidence that V1 has been validated as the current champion in the official snapshot.

Chapter D

Test and Contribution Matrix

D.1 Recommended validation commands

```
# Processed data (read-only unless --write-report is supplied)
python -m pipeline.verify_processed_data \
    --output-dir data/processed/ieee_cis_fraud_risk

# Model
cd model
uv run ruff check .
uv run pytest -q

# Backend
cd backend
uv run ruff check src tests
uv run pytest -q

# Frontend
cd frontend
npm run lint
npm run build
python scripts/check-ui.py
```

D.2 Acceptance scenarios

Table D.1: Acceptance scenarios

Scenario	Expected evidence
Processed handoff	Verifier passes every schema, row, split, and artifact check
Real model startup	<code>/meta</code> returns expected version with no warning
Full-feature score	<code>scoring_mode=full_feature</code> ; probability bounded
Partial score	<code>partial_demo</code> is visible; required fields enforced
Unseen category	Request succeeds with training-defined unknown mapping
SHAP absent	UI renders explicit unavailable state
Review update	Status/label persist and queue/statistics invalidate
CSV row error	Good rows commit; bad row appears in bounded error list
Job reconnect	Active job is returned after browser refresh
Model rollback	V1 loads and <code>/meta</code> reports V1

D.3 Team roster and repository role mapping

Table D.2: Group 1 roster and recorded subsystem contributions

Team member	Repository evidence
Dương Bình An	Data processing, EDA, feature engineering, contracts, and handover
Lê Quang Tuyển	Named team member; a subsystem-specific role is not stated in the reviewed repository
Phạm Đức Long	React dashboard, risk/SHAP presentation, review UX, and import UI
Đỗ Quốc Trung	FastAPI, SQL persistence, imports, background data loading, and Docker
Dương Hồng Quân	Model training, comparison, SHAP, artifact packaging, and scoring
Nguyễn Lê Hồng Nhi	Named team member; a subsystem-specific role is not stated in the reviewed repository

The full roster and supervisor were supplied by the project team. Detailed contribution statements should be confirmed by all members before submission.